

UNCOVERING CONSUMER PURCHASING PATTERNS: MARKET BASKET ANALYSIS ON BAKERY TRANSACTION DATA USING ASSOCIATION RULE MINING

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ABSTRACT

This report details the design and deployment of a data-driven retail intelligence system engineered to identify hidden consumer purchasing patterns within a high-volume bakery ecosystem. Utilizing transactional logs comprising over 20,000 recorded interactions, the research formalizes a multi-stage data processing and transformation framework to feed an unsupervised machine learning architecture. Association rule mining is executed via the classic Apriori framework to establish baseline operational trends across inventory assets.

Experimental evaluations successfully isolated statistically robust product associations by filtering mined outputs against specific Support, Confidence, and Lift thresholds. The resulting data structures revealed that high-volume anchor products like coffee act as recurring item transaction baselines, while complementary inventory elements like cakes and pastries generate the highest relational lift values. The extracted rules provide actionable business intelligence capable of optimizing brick-and-mortar shelf layouts, engineering strategic promotional item bundles, and refining supply-chain inventory preparations.

Index Terms— Market Basket Analysis, Association Rule Mining, Apriori Algorithm, Data Transformation, Unsupervised Learning, Retail Optimization.

1. INTRODUCTION

In the modern retail, hospitality, and commercial dining sectors, understanding consumer purchasing habits is an essential prerequisite for optimizing operational throughput and maximizing cross-selling revenue. Market Basket Analysis (MBA) stands as a foundational unsupervised data mining methodology that analyzes historical point-of-sale receipt logs to discover item-to-item co-purchase dependencies. By uncovering these implicit relational cross-purchases, store management can move away from speculative layout configurations and transition toward empirical, data-validated inventory strategies.

This term project establishes a comprehensive transaction processing and knowledge-discovery pipeline applied to an empirical bakery database. The processing model cleans transactional noise, restructures irregular human entry records, and maps textual lists into high-dimensional mathematical spaces appropriate for boolean evaluation. The primary goal is to isolate strong, reliable behavioral associations that can guide real-world business choices, such as strategic menu placements and cross-promotional bundles.

The remainder of this research paper is organized systematically into individual sections: Section 2 outlines the mathematical formulation of the target problem alongside verified project milestones. Section 3 covers the algorithmic mechanics of the matrix encoding layer and the down-closure candidate pruning steps of the Apriori framework. Section 4 presents the physical execution environment, exploratory data analysis visualizations, and primary numerical outputs. Section 5 breaks down the operational real-world value of the findings alongside core algorithmic scaling bottlenecks. Section 6 provides a summary of the framework's contributions and defines the development path for the upcoming benchmarking term project phase. Finally, Section 7 lists the links to the verified open-source digital artifacts.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Given an unstructured collection of point-of-sale transactions where an item checkout entry contains non-uniform naming formats, localized operational entries (such as administrative log balances or register corrections), and row-by-row item lists matching a common transaction receipt index, the objective is to model an end-to-end unsupervised analytical architecture. This system must parse the data stream without manual tagging, group item occurrences by checkout instance, map them into a structured boolean space, and compute precise association parameters.

The primary operational constraint is to extract rules that represent non-trivial purchasing behavior by isolating pairings that display high statistical dependencies rather than independent, random co-occurrences. The analytical model must screen out high-volume category noise (items bought frequently purely by baseline chance) and operate within a strict processing safety envelope where the extracted rule pool remains interpretable and free from combinatorial rule explosion.

2.2. Objectives

The structural milestones established to evaluate the success of this term project are defined as follows:

- **Data Refinement and Curation:** To implement structural string scrubbing filters and conditional exclusions to remove text spacing mismatches and non-purchase operational log rows.

- **Dimensional Space Reshaping:** To build a transactional encoding matrix that maps transaction item collections into a boolean vector matrix.
- **Frequent Pattern Mining:** To deploy the Apriori pruning framework using a minimum support coefficient threshold ($\text{min_support} = 0.01$) to filter out rare asset variations.
- **Statistical Association Extraction:** To generate and isolate actionable commercial rules by enforcing a strict lift threshold ($\text{lift} \geq 1.0$) to prove statistical dependency.
- **Open-Source Portfolio Architecture:** To package all functional scripts and code notebooks inside a public, verifiable repository.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

The systematic operational flow of the data mining system follows a sequential pipeline, running from raw database inputs to the extraction of actionable business intelligence.

Functional pipeline of the Market Basket Analysis architecture, demonstrating data progression from initial ingestion to rule extraction.

The dataset is first ingested into a tabular memory framework where alphanumeric structural normalization is applied. Next, individual transaction items are aggregated by their receipt indices and mapped into a high-dimensional sparse binary grid. This one-hot encoded matrix is then passed to the frequent pattern extraction layer, which applies the down-closure rule to eliminate infrequent candidate items. Finally, the generated rules are filtered across multi-dimensional support, confidence, and lift thresholds to populate the final business intelligence table.

3.2. Technical Details and Formulations

Let $I = \{i_1, i_2, \dots, i_m\}$ represent the universal set of unique distinct items available within the bakery inventory, and let $T = \{t_1, t_2, \dots, t_n\}$ represent the comprehensive database of customer checkouts, where each individual transaction t_k is an itemset such that $t_k \subseteq I$.

3.2.1. Boolean Matrix Reshaping

To enable fast vector matrix evaluations, the sequential transaction text records are converted into a binary matrix M of dimensions $n \times m$. The allocation of values within this matrix is formally defined as:

$$M_{k,j} = \begin{cases} 1 & \text{if } i_j \in t_k \\ 0 & \text{if } i_j \notin t_k \end{cases} \quad (1)$$

This binary representation allows the mining engine to calculate support levels via rapid, column-wise bitwise intersections rather than executing expensive text searches across the raw database.

3.2.2. Algorithmic Pruning and Association Metrics

An association rule is defined as an implication expression of the form $A \rightarrow B$, where $A \subset I$, $B \subset I$, and $A \cap B = \emptyset$. The itemset A is termed the antecedent, while B is termed the consequent. The engine filters candidates by sequentially computing three core mathematical parameters:

Support: Measures the baseline popularity of an itemset within the complete transaction log space. It estimates the joint probability $P(A \cup B)$:

$$\text{Support}(A \rightarrow B) = \frac{|\{t_k \in T \mid (A \cup B) \subseteq t_k\}|}{n} \quad (2)$$

Confidence: Quantifies the conditional certainty of the rule, estimating the conditional probability $P(B \mid A)$ of a purchase occurring given the presence of the antecedent:

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} \quad (3)$$

Lift: Evaluates the statistical dependency and strength of the rule by calculating the ratio of the observed joint support to the expected support if the antecedent and consequent were completely independent:

$$\text{Lift}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A) \times \text{Support}(B)} \quad (4)$$

The model uses the down-closure property of support: if an itemset is infrequent at a given length, all of its supersets will also be infrequent, allowing them to be safely pruned without further database passes.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

The model was built inside a local project workspace using Visual Studio Code. The software architecture was isolated within a dedicated Python virtual environment running Python 3.11, relying on the `pandas` ecosystem for data manipulation, `matplotlib` and `seaborn` for chart rendering, and the `mlxtend` library for frequent pattern operations.

The source dataset, `bread basket.csv`, tracks real-world transactions from a commercial bakery. After dropping operational administrative logs, the final clean dataset consists of 20,507 rows spanning 9,465 unique multi-item checkouts and 94 distinct product categories. The frequent pattern engine was initialized with a minimum support floor ($\text{min_support} = 0.01$) and a minimum lift threshold ($\text{min_lift} = 1.0$).

4.2. Results and Analytical Findings

The modeling pipeline successfully extracted a set of actionable business rules. Table 1 details the top associations discovered by the pipeline, sorted in descending order of their Lift scores.

Table 1. Top Extracted Association Rules (Apriori Framework Baseline)

Rank	Antecedent (A)	Consequent (B)	Support	Conf.	Lift
1	{cake}	{tea}	0.0237	0.2288	1.6047
2	{medialuna}	{coffee}	0.0351	0.5692	1.1898
3	{pastry}	{coffee}	0.0475	0.5521	1.1516
4	{sandwich}	{coffee}	0.0382	0.5323	1.1153
5	{cake}	{coffee}	0.0547	0.5269	1.1091

Exploratory data visualizations highlighted that high-volume products, such as coffee (appearing in 5,471 rows) and bread (appearing in 3,325 rows), establish a dominant baseline across the transactional distribution. When analyzing metrics via the multi-dimensional distribution scatter plot, it was observed that rules leading to a coffee consequent consistently show high confidence scores (exceeding 50%). However, their lift scores remain close to 1.1, indicating that these high confidence values are largely driven by coffee’s baseline popularity rather than a unique cross-purchasing affinity. Conversely, specialized pairings like cake and tea display lower absolute support but significantly higher lift coefficients, pointing to genuine cross-product affinity.

5. DISCUSSION

The rules extracted by the model offer clear, actionable business intelligence for brick-and-mortar retail optimization. The association rule $\{cake\} \rightarrow \{tea\}$ yields the highest lift score (1.6047), proving that a customer purchasing a cake is 1.6 times more likely to add tea to their order compared to a standard baseline shopper. Similarly, high-volume rules like $\{pastry\} \rightarrow \{coffee\}$ show an exceptional confidence level of 55.21%.

From an operational standpoint, management can leverage these insights to optimize store layouts and drive cross-selling revenue. For instance, high-lift items (such as cakes and pastries) can be placed adjacent to high-volume anchors (such as tea and coffee counters) to prompt impulse buying. Additionally, these metrics can guide the creation of targeted breakfast combinations or automated “Buy One, Get One” discount pairings during slower operational hours.

5.1. Technical Limitations and Bottlenecks

Despite successfully mining the target dataset, the baseline Apriori framework faces significant architectural bottlenecks when scaling. The algorithm relies on an iterative, level-by-level candidate generation-and-test process. To find frequent itemsets of length k , it must repeatedly scan the entire transaction database to compute support counts.

As inventory size or transaction volume scales up, the number of potential candidate combinations grows exponentially. This combinatorial explosion creates severe memory overhead and heavy processing bottlenecks, making the standard Apriori framework inefficient for massive enterprise retail datasets.

6. CONCLUSION AND FUTURE WORK

This term project successfully models and deploys a complete Market Basket Analysis framework, transforming raw point-of-sale bakery records into structured business intelligence. By mapping transaction rows into a boolean matrix, the pipeline systematically cleaned, transformed, and mined hidden consumer purchasing habits. The project achieved all structural objectives, establishing an automated pipeline that isolates statistically valid item affinities while filtering out high-volume category noise.

To address the performance limitations identified during this phase, the next term project submission course (DA378) will pivot toward an advanced architectural benchmarking study. The upcoming phase will implement and evaluate the ****FP-Growth (Frequent Pattern Growth) Algorithm****. Unlike Apriori, the FP-Growth framework avoids candidate generation entirely by compressing the

transaction database into a highly efficient tree structure (an FP-Tree) and scanning the raw database only twice. Future work will feature execution runtime performance charts and memory footprint comparisons across scaling support thresholds to provide deep engineering analysis.

7. ARTIFACTS AND DEMONSTRATIONS

To ensure reproducibility and satisfy the verification requirements of the online degree platform, all source files are packaged and hosted openly[cite: 377]:

- **Code Portfolio Repository:** The complete, functional Python notebook file, dependency specifications (`requirements.txt`), and the primary source dataset (`bread_basket.csv`) are publicly hosted on GitHub at: <https://github.com/Arun8k/Market-Basket-Analysis-Bakery>[cite: 263, 378].
- **Recorded Video Demonstration:** A detailed 10-minute presentation covering the source code walkthrough, data transformation steps, and result visualizations is published on YouTube at: [YOUR_YOUTUBE_VIDEO_LINK_HERE](#)[cite: 137, 379].

8. REFERENCES